PROJECT REPORT

Swiggy Restaurant Recommendation System using Cosine Similarity

1. Introduction

This project builds a restaurant recommendation engine using cosine similarity and a Streamlit web interface.

2. Objective

To recommend restaurants based on city, cuisine, cost, and rating preferences.

3. Data Cleaning

Duplicate removal, missing value handling, text normalization, and cleaned_data.csv creation.

4. Preprocessing

One-Hot Encoding applied to city and cuisine. Encoded dataset and encoder.pkl created.

5. Recommendation Methodology

Cosine similarity measures similarity between user preference vector and restaurant vectors. Top matches are selected.

6. Streamlit Application

Interactive UI allowing users to filter preferences and view recommended restaurants.

7. Results

- Accurate recommendations

- Real-time similarity computation

- Clean and intuitive interface

8. Conclusion

A fully functional pipeline built for restaurant recommendations using cosine similarity.

9. Future Scope

TF-IDF menu similarity, embeddings, hybrid models, and geolocation filtering.